

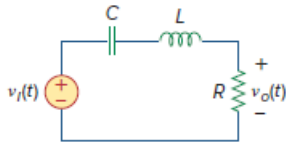
Problem

Realize the function

$$G(s) = \frac{V_o(s)}{V_i(s)} = \frac{4s}{s^2 + 4s + 20}$$

using the circuit in Fig. 16.31. Select $R = 2 \, \Omega$, and determine L and C .

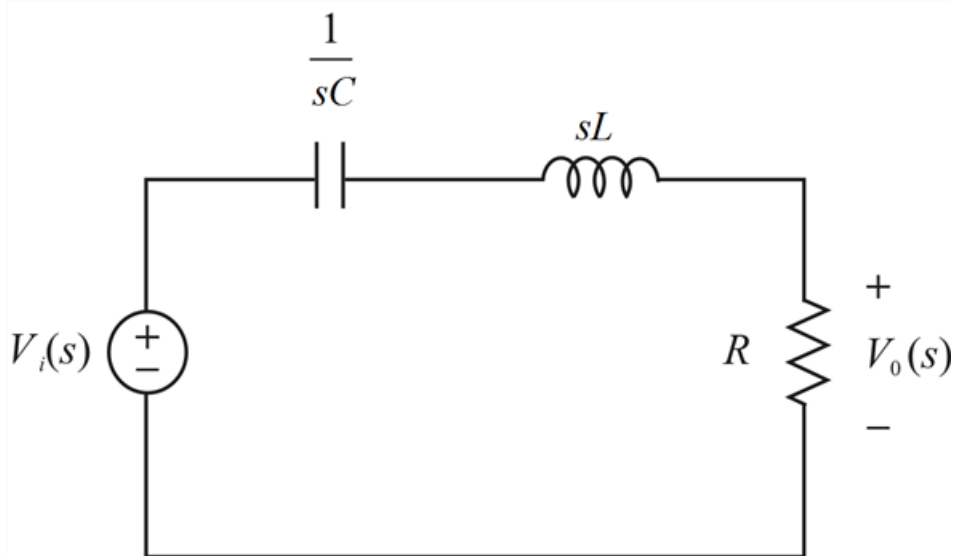
Figure 16.31:



Step-by-step solution

Step 1 of 3

The circuit in s-domain is



Step 2 of 3

Using voltage division rule, then

$$V_o(s) = \frac{R}{R + sL + \frac{1}{sC}} V_i(s)$$

$$\frac{V_o(s)}{V_i(s)} = \frac{RsC}{s^2 LC + sRC + 1}$$

$$H(s) = \frac{sRC}{LC \left(s^2 + \frac{sRC}{LC} + \frac{1}{LC} \right)}$$

$$H(s) = \frac{s \frac{R}{L}}{s^2 + s \frac{R}{L} + \frac{1}{LC}} \quad \text{--- (1)}$$

Given transfer function is

$$H(s) = \frac{4s}{s^2 + 4s + 20}$$

Compare given transfer function with equation (1) then

$$\frac{R}{L} = 4, \frac{1}{LC} = 20$$

Select $R = 2\Omega$

$$\frac{2}{L} = 4$$

$$L = \frac{1}{2}$$

$$\therefore \boxed{0.5 \text{ H}}$$

$$\frac{1}{LC} = 20$$

$$\frac{1}{(0.5)C} = 20$$

$$\frac{1}{C} = (20)(0.5)$$

$$\therefore \boxed{C = 0.1 \text{ F}}$$